

DisasTRACE: Emergency Incident Reporting & Dispatch Platform

Built for CDRRMO Baliwag City & Public Assistance and Command Center (PACC)
Full System Architecture, End-to-End Operational Flows, Data Models, API Registry, and Security.
Includes 1km PostGIS Spatial Dispatching, Realtime WebSockets, Touch E-Signatures, & Offline Sync.

1. EXECUTIVE SUMMARY & SYSTEM BACKGROUND

A. PROBLEM STATEMENT & BACKGROUND

- **Traditional Emergency Response Challenges:** CDRRMO Baliwag City previously relied on manual phone hotlines and paper-based logs. This led to verbal address miscommunications, delayed responder dispatching, lack of real-time responder location visibility, and vulnerable paper record keeping.
- **Solution Overview:** DisasTRACE is a centralized, multi-platform emergency management platform. It connects Public Residents, Ambulance Responders, PACC Dispatchers, and CDRRMO Executive Admins through a high-performance Android mobile app and a Next.js real-time web dashboard.

B. CORE MISSIONS & GOALS

- **Eliminate Call Bottlenecks:** Allow residents to submit digital SOS requests tagged with exact GPS coordinates and camera photo evidence.
- **Automated 1km Proximity Selection:** Use PostGIS geospatial indexing (ST_DWithin) to automatically find and dispatch the nearest available ON_DUTY ambulance within a 1.0km radius.
- **Real-Time GPS Telemetry:** Broadcast responder GPS positions, speed, and heading to PACC Web Command Center via Supabase Realtime WebSockets.
- **Paperless Electronic Forms:** Digitally capture Pre-Hospital Care Reports (PCR) and Driver's Trip Tickets (DTT) with touch finger E-Signatures and instant PDF exports.

2. STAKEHOLDER PERSONAS & SYSTEM ROLES

1. **Resident / Public User (Android App):** Registers with phone OTP verification and government ID. Submits emergency SOS requests.
2. **Ambulance Responder (Android App):** Receives timed 30-second dispatch offer drawers with resident photo evidence and location.
3. **PACC Dispatcher / Admin (Next.js Web):** Monitors incoming emergency requests, triages incidents (VERIFICATION_PENDING).
4. **CDRRMO Super Admin (Next.js Web):** Reviews user/responder registrations, manages system accounts, views executive reports.

3. END-TO-END OPERATIONAL WORKFLOWS

A. ACCOUNT REGISTRATION & VERIFICATION GATE

- **OTP SMS Verification:** Users register with phone verification powered by textbee.dev SMS gateway.
- **Government ID Upload:** Mobile registrants upload a photo of a government-issued ID stored in Supabase Storage (ids/{userId}).
- **Super Admin Verification Gate:** Accounts enter PENDING status. All mobile app screens are locked until a CDRRMO Super Admin verifies the account. If rejected, an SMS alert is sent with rejection reasons.

B. RESIDENT EMERGENCY SOS FLOW

- **GPS Tagging & Photo Capture:** Residents tap SOS, automatically locking GPS coordinates (latitude/longitude) and taking mandatory incident scene photos.
- **Structured Classification:** Captures call nature, emergency type, estimated victims, and chief complaint details.
- **Live Responder Tracking:** Post-dispatch, residents view assigned ambulance movement on an interactive MapLibre map with dynamic ETA updates.

C. PACC TRIAGE & VERIFICATION QUEUE

- Verification Queue Landing: Emergency requests enter the PACC Queue as VERIFICATION_PENDING.
- Verification Action: Dispatchers inspect incident details, verify authenticity (VERIFIED), and choose either automated spatial dispatch or manual responder selection.

D. AUTO-DISPATCH ENGINE & 1KM PROXIMITY CASCADE

- PostGIS 1km Spatial Search: Executes a spatial query using ST_DWithin(location_geom, ST_SetSRID(ST_MakePoint(lng, lat), 4326), 1000) to find nearest ON_DUTY responders within 1.0km.
- 30-Second Timed Offer: Transmits a 30-second offer (currentOfferResponderId) to the candidate. Candidate's duty status becomes ACTIVE_DISPATCH.
- Automated Cascading: If the responder declines or the 30-second timer expires, the engine automatically cascades to the next nearest candidate within 1km.
- Fallback Escalation: If no candidate accepts within 1km, the incident transitions to PACC_MANUAL status and escalates back to the PACC Dispatch Queue for manual assignment.

E. RESPONDER MOBILE OPERATIONS & DIGITAL SIGNATURES

- Offer Accept/Decline Drawer: Responders view timed offer cards with resident scene photos, incident location, and distance.
- Telemetry Broadcast: Live GPS location, heading orientation, and speed (kph) are broadcast to the command center via WebSockets (useBroadcastTracker).
- Turn-by-Turn Navigation & Hospital Search: Interactive route navigation to scene and receiving hospital.
- Touch E-Signatures (SignaturePadModal): Touch finger canvas allowing on-screen signature drawing for Patient/Guardian, Witness, Responder, Driver, and Authorized Representative.
- Pre-Hospital Care Report (PCR) & Driver's Trip Ticket (DTT): Electronic forms capturing vitals, GCS points, SAMPLE history, narrative reports, and vehicle fuel controls.

F. CDRRMO EXECUTIVE DASHBOARD & PDF EXPORTS

- Command Center Interactive Map: Displays live active incidents, hospital locations, and real-time ambulance fleet positions.
- KPI Analytics: Dynamic charts (Recharts) visualizing daily incidents, average response times, severity breakdown, and responder activity logs.
- Report Inspection & Vector PDF Export: Report inspection modal rendering SVG vector signature cards, with one-click export (exportPatientCareReportPDF & exportDriverTripTicketPDF) converting vector paths to high-res PNG images via HTML5 Canvas (convertSvgPathToDataUrl).

G. OFFLINE DRAFT PERSISTENCE & 5-MINUTE RECURRING REMINDER LOOP

- Local File Caching: PCR and Trip Ticket drafts created offline are saved in device storage (disas_trace_drafts.json).
- 5-Minute System Reminder: When connection recovers (isOnline === true), the app triggers high-priority notifications and haptics every 5 minutes reminding responders to submit pending drafts.
- Background Queue Replay: Sequentially flushes local offline queue actions to backend API endpoints upon connection return.

4. DETAILED TECHNOLOGY STACK SPECIFICATIONS

- Web Front-End: Next.js 16 (React 19, App Router), TypeScript, Tailwind CSS v4, Lucide Icons, Shadcn UI, Recharts.
- Mobile Front-End: React Native with Expo (SDK 52, Expo Router), NativeWind / Tailwind CSS, @maplibre/maplibre-react-native (Android APK).
- Backend & Serverless API: Next.js API Routes (Node.js Serverless runtime), Supabase Cloud Platform.
- Database & Spatial Extension: Supabase PostgreSQL DBMS with PostGIS spatial extension (locationGeom spatial index, ST_DWithin 1km radius query, ST_SetSRID).
- ORM Layer: Drizzle ORM (type-safe SQL schema & parameterized query builder).

- **Authentication & Security:** Supabase Auth with Argon2id / bcrypt password hashing, JWT claims, 4-tier RBAC (pacc_admin, cdrrmo_admin, ambulance_responder, resident), and PostgreSQL Row Level Security (RLS).
- **File & Asset Storage:** Supabase Storage Buckets (ids/, scenes/, exports/).
- **Mapping & Tile Engine:** OpenFreeMap / MapLibre GL, OpenStreetMap (OSM) vector tiles (Zero-cost, no API key).
- **PDF Generation:** jsPDF & jsPDF-AutoTable with HTML5 Canvas vector-to-PNG renderer (convertSvgPathToDataUri).

5. DATABASE SCHEMA & API ENDPOINT REGISTRY

A. DATABASE TABLES (POSTGRESQL / DRIZZLE ORM)

- **public.users:** Id, fullName, phone, role, dutyStatus, verificationStatus, lastLatitude, lastLongitude, locationGeom (PostGIS point).
- **public.verification_requests:** Id, residentId, locationDescription, type, latitude, longitude, imageUrl, status, createdAt.
- **public.incidents:** Id, requestId, responderId, status, assignedAmbulance, currentOfferResponderId, offerExpiresAt, dispatchMethod, createdAt.
- **public.patient_care_reports:** Id, incidentId, patientName, patientAge, dispatchInfo, emergencyType, initialAssessment, vitalsLogs, sampleHistory, narrativeReport, handoffSignatures, liabilityRelease, respondingTeam.
- **public.driver_trip_tickets:** Id, incidentId, driverName, vehiclePlate, passengerName, placesVisited, tripLog, gasolineConsumed, lubricants, speedometer, remarks, signatures.

B. REST API ENDPOINTS REGISTRY

- **POST /api/auth/register** — Mobile account registration with OTP verification.
- **POST /api/verification-requests** — Submit resident SOS help request with photo and GPS.
- **POST /api/verification/[id]/dispatch** — Manual responder dispatch by PACC dispatcher.
- **POST /api/incidents/respond** — Responder accept/decline dispatch offer.
- **POST /api/incidents/status** — Incident status updates (EN_ROUTE, ON_SCENE, TO_HOSPITAL, RESOLVED).
- **POST /api/responder/location** — Live GPS telemetry update writing PostGIS locationGeom.
- **POST /api/reports** — Submit electronic PCR and Driver's Trip Ticket payloads.

6. SECURITY, COMPLIANCE & DISASTER RECOVERY

- **100% SQL Injection Protection:** All database interactions utilize Drizzle ORM parameterized queries.
- **Strict Role-Based Access Control (RBAC):** Enforced at client navigation guards and server-side API authorization headers.
- **Row Level Security (RLS):** Supabase PostgreSQL tables are protected by strict RLS policies.
- **Data Encryption:** HTTPS / TLS 1.3 encryption for data in transit; AES-256 for storage assets.
- **Automated Cloud Backups:** Daily automated database backups with WAL Point-in-Time Recovery (PITR).
- **Immutable Audit Trail:** Every incident status transition, dispatch offer, acceptance, decline, and telemetry log is stored in audit logs.